Paper review · Archeologist

Where FAST came from, and what it led to

FAST: Efficient Action Tokenization for Vision-Language-Action Models. Pertsch, Stachowicz, Ichter, Driess, Nair, Vuong, Mees, Finn and Levine. Physical Intelligence, UC Berkeley and Stanford.

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arXiv:2501.09747 · January 2025
Pavel Bushuyeu · October 2026
bushuyeu.com/review-fast-lineage

FAST in one slide

The problem

RT-2 and OpenVLA turn each action dimension at each time step into one of 256 bin tokens. At high control rates neighbouring tokens are almost the same, so the model learns to copy the last token.

The fix

Compress each one-second action chunk before tokenizing: a discrete cosine transform per dimension, scale and round as JPEG does, then byte-pair encoding.

π0-FAST matches diffusion π0 with **5× fewer GPU hours** of training, but needs about **750 ms** per action chunk at inference against about 100 ms.

Dataset	Rate	Binning	FAST	Ratio
BridgeV2	5 Hz	35	20	1.75×
DROID	15 Hz	105	29	3.6×
Table bussing	20 Hz	140	28	5.0×
Shirt folding	50 Hz	700	53	13.2×

Tokens per one-second chunk at similar reconstruction error. FAST, Table I.

The FAST tokenizer

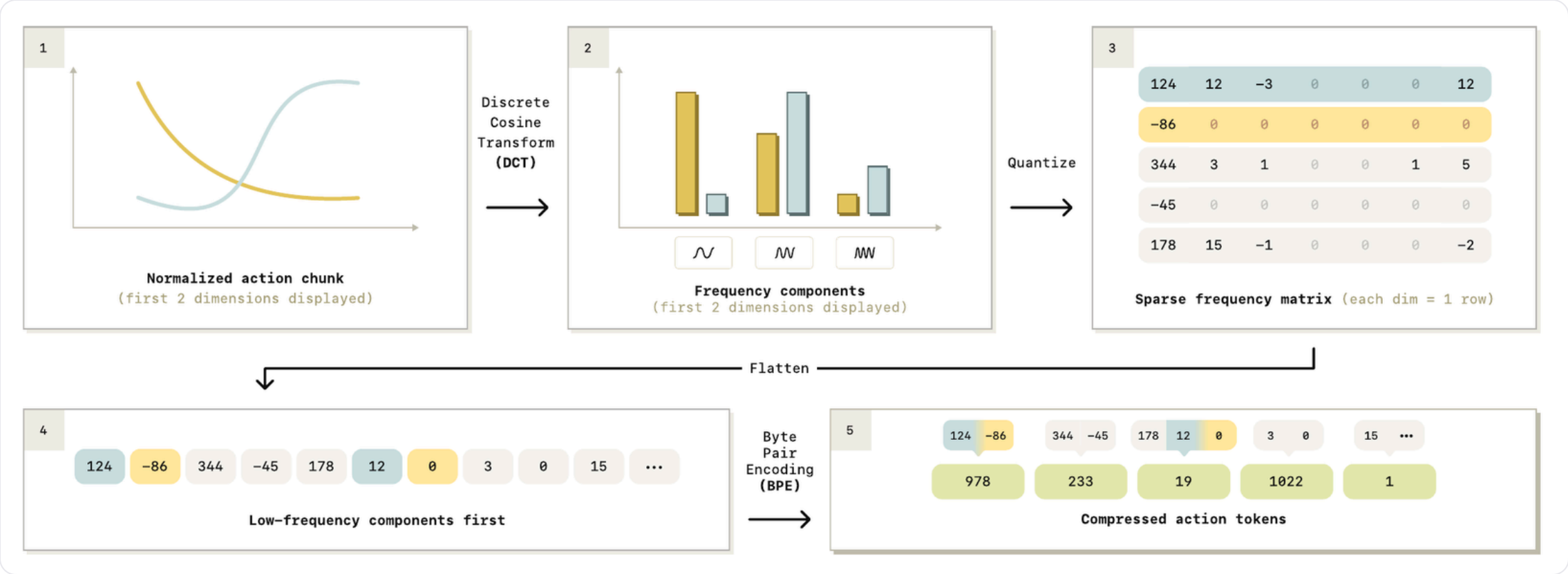
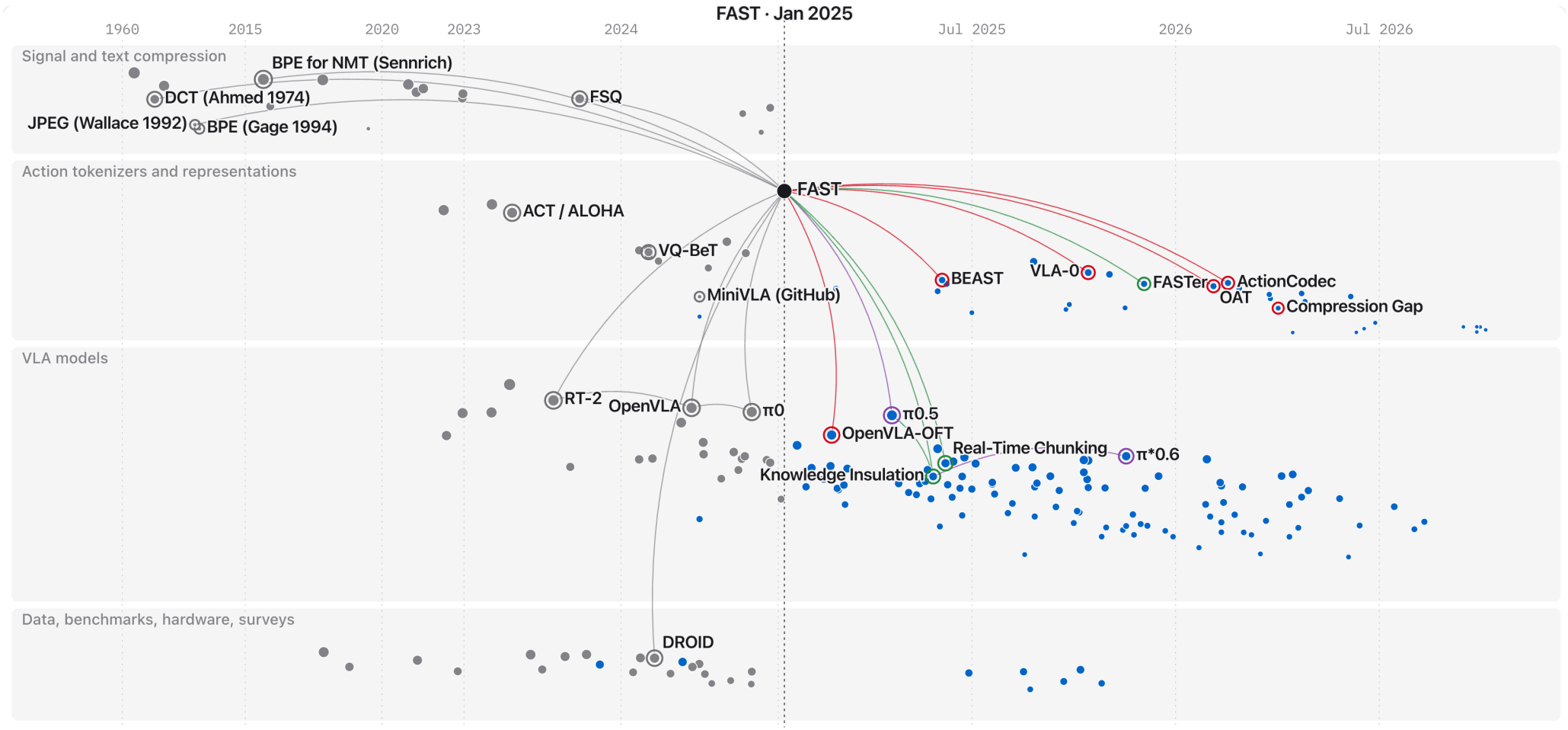


Fig. 4 of FAST: DCT per action dimension, quantization to a sparse matrix, flattening with the low frequencies first, then byte-pair encoding. Pertsch et al., arXiv:2501.09747, CC BY 4.0.

Citation timeline



References left of January 2025, citing papers right; higher means more cited; rings mark the papers in this talk. Semantic Scholar, 19 Oct 2026.

Prior works

1974–2016	**DCT, JPEG and byte-pair encoding** The compression idea, from outside robotics: a frequency transform, rounding, then BPE.
2017–2024	**VQ-VAE, FSQ, VQ-BeT, MiniVLA** The road not taken: learned quantizers work for coarse reconstruction, fail on high-frequency control.
Apr 2023	**ACT and ALOHA** Action chunking: predict a second of actions at once.
Jul 2023	**RT-2** The first VLA: one bin token per action dimension per time step.
Mar 2024	**DROID** Binning-based VLAs trained poorly on it; π0-FAST works zero-shot in new scenes.
Jun 2024	**OpenVLA** An open 7B VLA that kept binning; FAST's main baseline.
Oct 2024	**π0** The same lab's earlier model: FAST's base and its point of comparison.

OpenVLA and RT-2, then π0

Binning, kept and replaced

- FAST keeps their recipe: a VLM fine-tuned to write action tokens, trained with next-token prediction.
- It changes only the tokenizer. Its baseline is labelled “OpenVLA-style” (Fig. 2).
- Karl Pertsch, FAST's first author, co-wrote RT-2, OpenVLA, Octo and DROID: FAST fixes a limit of his own earlier models.

π0: same lab, the base model

π0 is PaliGemma-3B with a flow-matching action expert. π0-FAST keeps the backbone, drops the expert and writes FAST tokens into the VLM's vocabulary.

The paper compares the two decoders on one model: about 5× less training compute for π0-FAST, about 750 ms against 100 ms per chunk at inference.

What came after

Apr 2025	**π0.5** • Validates	Pre-trains on FAST tokens for all robot data; flow matching produces the actions. 8 of FAST's 9 authors.
May 2025	**Knowledge Insulation** • Extends	FAST tokens and a flow-matching expert, trained together with a stop-gradient.
Jun 2025	**Real-time chunking** • Extends	Removes the pause while the next action chunk is generated.
Nov 2025	**π*0.6** • Validates	Still predicts FAST tokens during training, through Knowledge Insulation.
Dec 2025	**FASTer** • Extends	A learned VQ tokenizer that treats an action chunk as an image.

Knowledge Insulation does FAST's future work

> "...testing the combination of compression-based action encodings with non-autoregressive decoding approaches like diffusion..."
>
> FAST, discussion of future work

- **Knowledge Insulation** (May 2025; Driess, Pertsch, Levine and others): the VLM learns from FAST tokens, a flow-matching expert produces the actions, and a stop-gradient keeps the expert from damaging the VLM.
- Train fast like π0-FAST, run fast like π0. π0.5 and π*0.6 use the same recipe.

FAST against its challengers

Paper	Objection to FAST	Uses instead	FAST against it, in that paper's setup
OpenVLA-OFT Feb 2025	About 750 ms between action chunks is too slow for high-frequency bimanual robots.	Continuous actions, parallel decoding and an L1 loss.	LIBERO, fine-tuned: π0-FAST **85.5%**, π0 94.2%, OpenVLA-OFT **97.1%**.
BEAST Jun 2025	Chunks of one length give token sequences of different lengths, which blocks parallel decoding.	Fixed-length B-spline tokens, no tokenizer training.	Same Florence-2, no robot pre-training: matches FAST on LIBERO-Long, beats it on CALVIN ABC→D.
VLA-0 Oct 2025	Custom tokenizers “often require intricate training pipelines”.	Actions written as plain-text numbers.	LIBERO: π0-FAST **86.0%** with robot pre-training, 71.8% without; VLA-0 **94.7%** without.

Each paper's own setup: compare within a row, not across rows. LIBERO is the average success over its four suites.

FAST against its challengers

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ActionCodec Feb 2026	"Fixed geometric priors" limit transfer across robots; decoded length must match exactly, so they pad or truncate.	A learned tokenizer designed from VLA training behaviour.	Same SmolVLM2-2.2B, tokenizer swapped: FAST **90.6%**, best of five existing tokenizers; ActionCodec **95.5%**.
OAT Feb 2026	Not every token sequence decodes; invalid ones make the policy "unstable, overly aggressive".	Learned, ordered tokens; any prefix decodes.	Small policies from scratch, LIBERO: FAST **23.0%**, diffusion policy 36.6%, OAT **56.3%**.
The Compression Gap Apr 2026	A fixed-capacity discrete code caps what a better vision encoder can add.	An analysis; continuous actions avoid the cap.	Not tested on FAST; shown for OAT and diffusion policy.

Each paper's own setup: compare within a row, not across rows. LIBERO is the average success over its four suites.

What the comparison shows

- **As a tokenizer, FAST holds up.** With only the tokenizer swapped it beats binning and VQ tokenizers (ActionCodec); BEAST only matches it on LIBERO-Long.
- **Decoding is the common objection.** Sequences that do not decode (OAT, ActionCodec), the delay per chunk (OpenVLA-OFT), variable length against parallel decoding (BEAST).
- **It needs robot pre-training.** Without it π0-FAST drops from 86.0% to 71.8% on LIBERO (VLA-0) and learns late (BEAST).
- **Its own lab answered the speed objection differently.** π0.5, Knowledge Insulation and π*0.6 train on FAST tokens and decode with flow matching.

OpenVLA-OFT

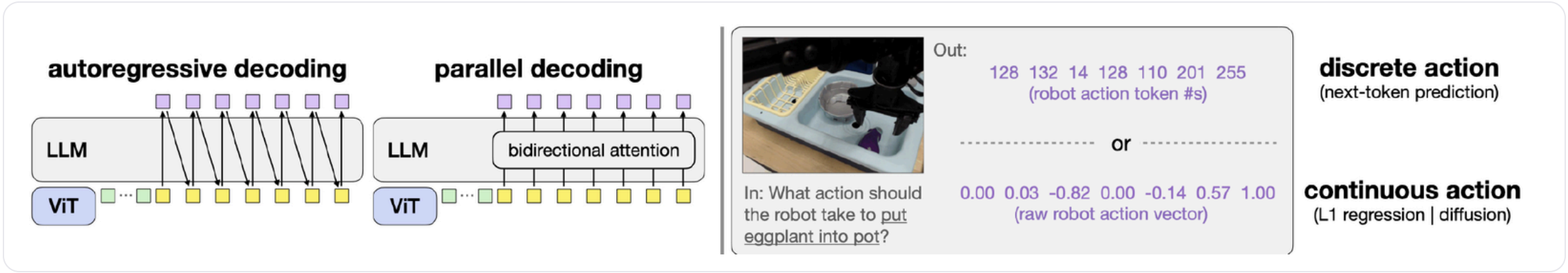
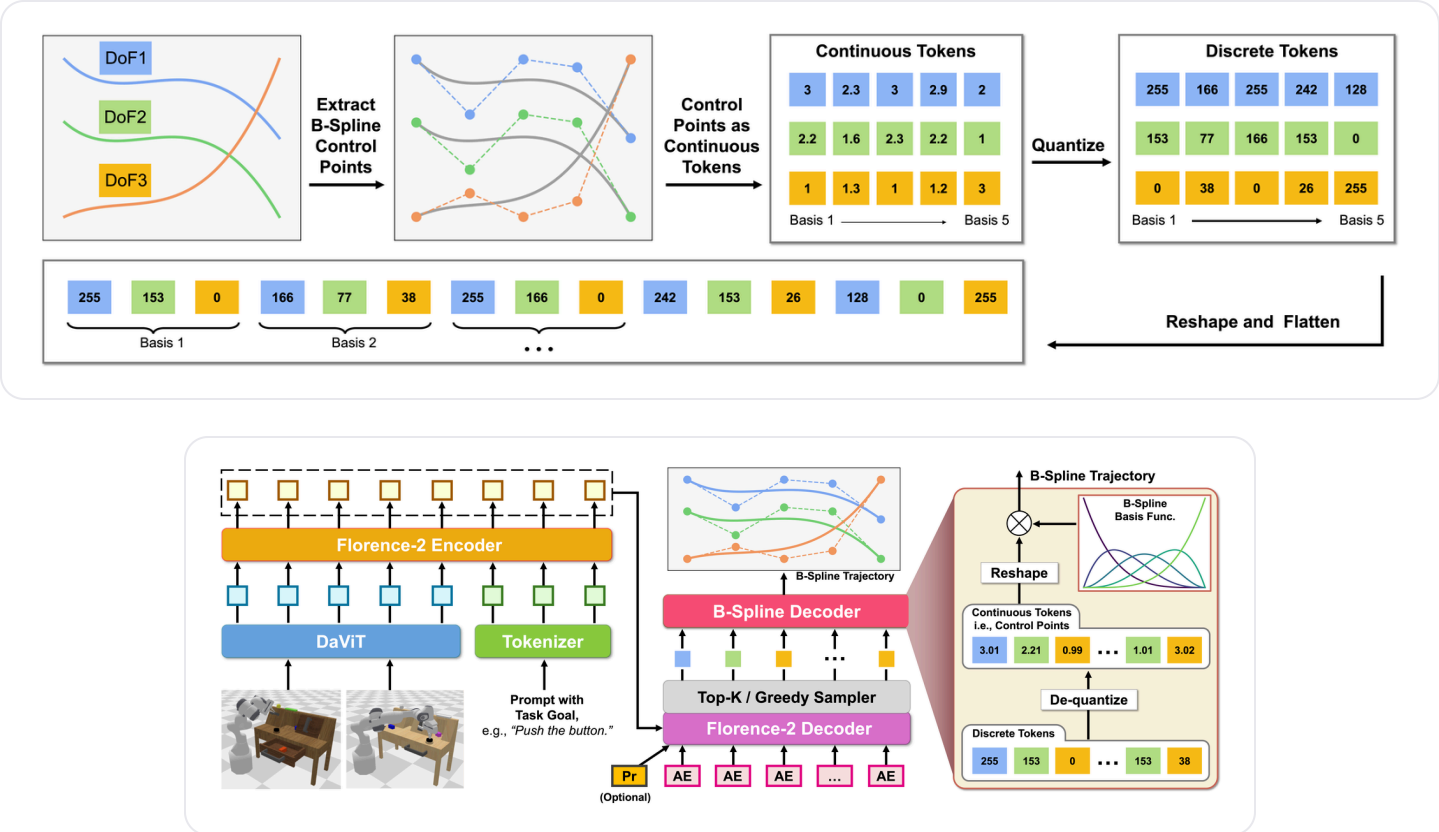


Fig. 2 of OpenVLA-OFT: autoregressive against parallel decoding (left), discrete action tokens against continuous actions (right). Kim, Finn and Liang, arXiv:2502.19645, CC BY 4.0.

BEAST



Figs. 2 and 3 of BEAST: the encoding pipeline (top) and BEAST-F, BEAST with Florence-2, where tokens of uniform length allow parallel decoding (bottom). Zhou et al., arXiv:2506.06072, CC BY-NC-SA 4.0.

VLA-0

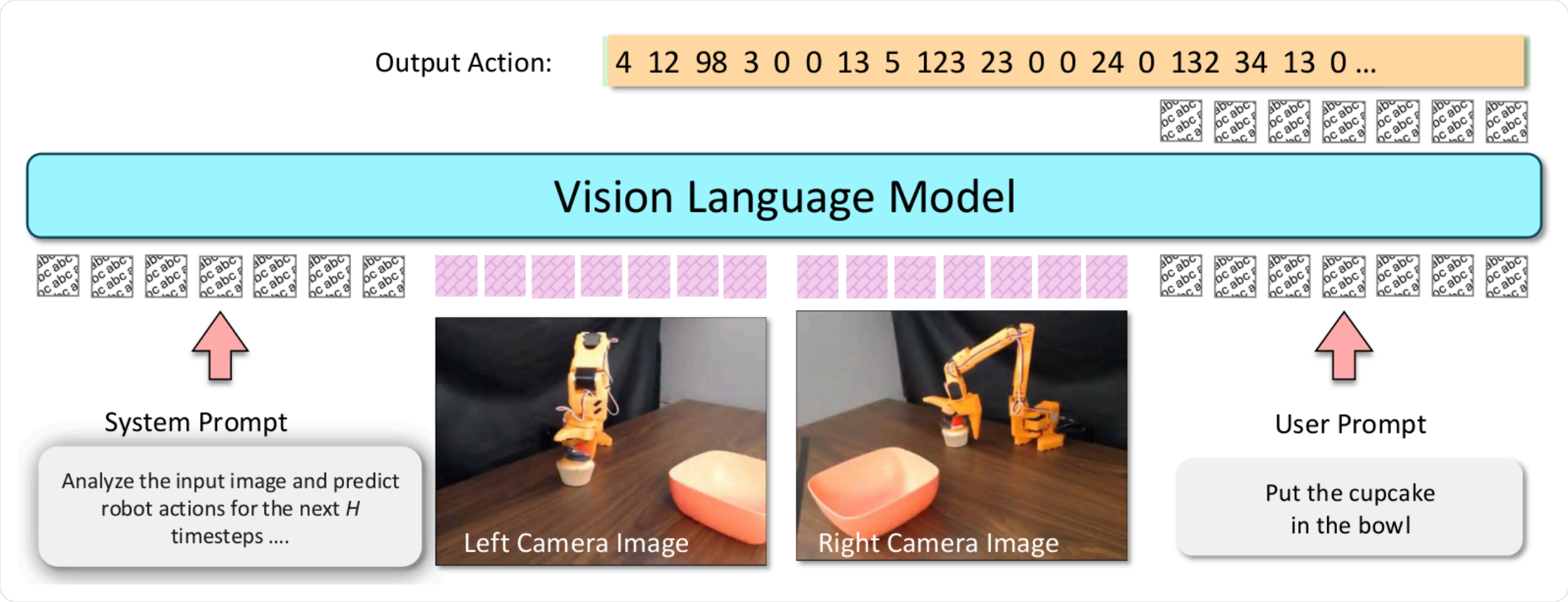


Fig. 3 of VLA-0: the VLM is not changed; it writes actions as space-separated integers. Goyal et al., arXiv:2510.13054, arXiv non-exclusive license; shown for commentary.

ActionCodec

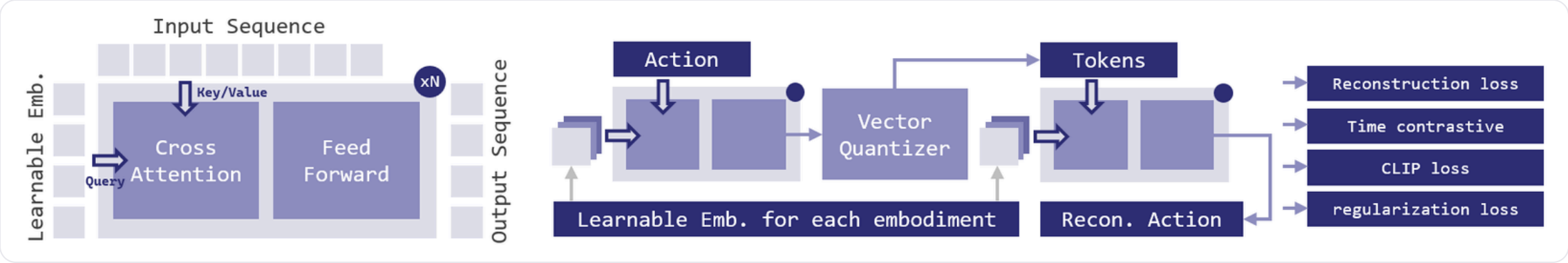


Fig. 2 of ActionCodec: a Perceiver-like encoder and decoder around a vector quantizer, with an embedding per embodiment and four training losses. Dong et al., arXiv:2602.15397, CC BY 4.0.

OAT

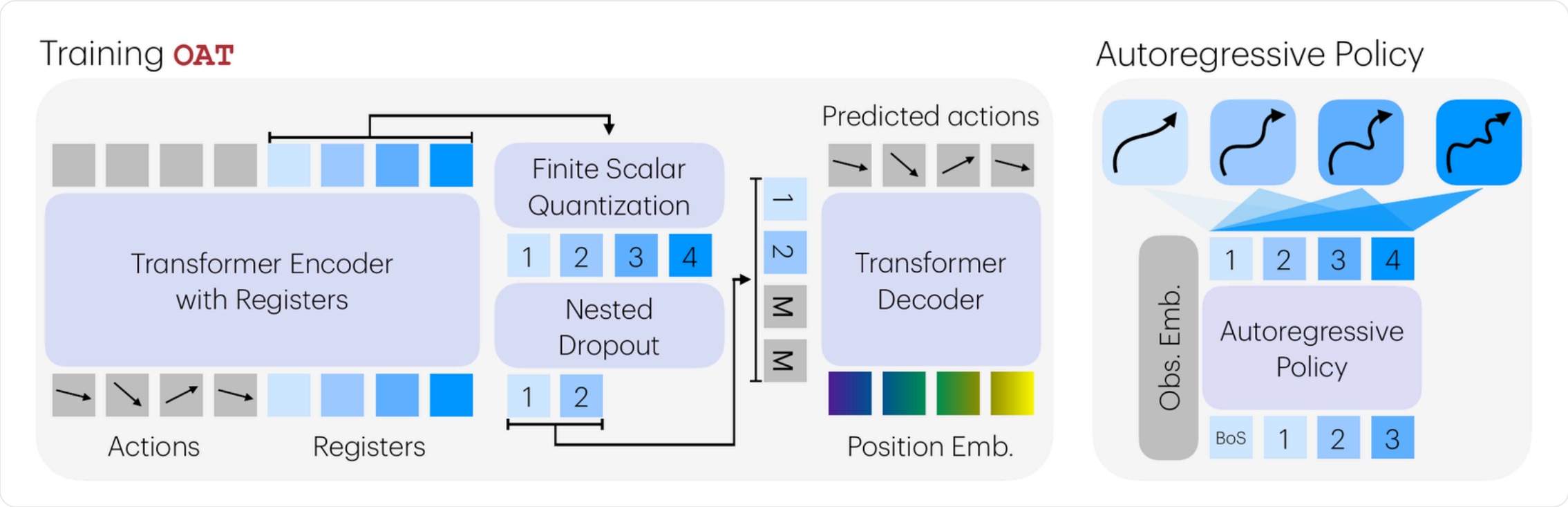


Fig. 3 of OAT: ordered tokens from a transformer encoder with registers, FSQ and nested dropout (left), generated by an autoregressive policy (right). Liu et al., arXiv:2602.04215, CC BY 4.0.

The Compression Gap

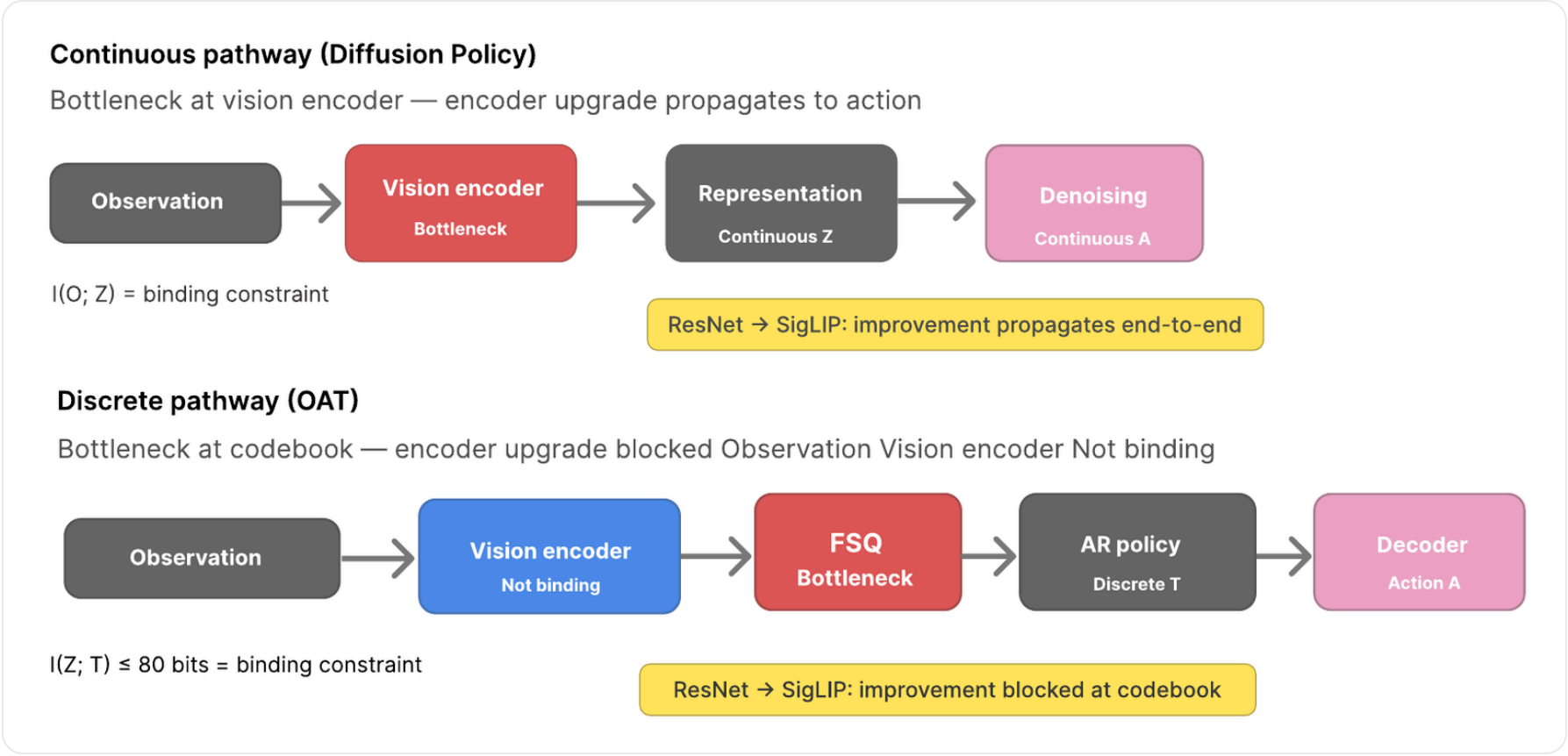


Fig. 1 of The Compression Gap: the vision encoder is the bottleneck with continuous actions, the codebook with discrete tokens. Shiba, arXiv:2604.03191, CC BY 4.0.

Sources

1. Pertsch et al. FAST: Efficient Action Tokenization for Vision-Language-Action Models. arXiv:2501.09747, 2025.
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7. Driess et al. Knowledge Insulating VLA Models. arXiv:2505.23705, 2025.
8. Kim, Finn and Liang. OpenVLA-OFT. arXiv:2502.19645, 2025.
9. Zhou et al. BEAST. arXiv:2506.06072, 2025.
10. Goyal et al. VLA-0. arXiv:2510.13054, 2025.
11. Dong et al. ActionCodec. arXiv:2602.15397, 2026.
12. Liu et al. OAT: Ordered Action Tokenization. arXiv:2602.04215, 2026.
13. Shiba. The Compression Gap. arXiv:2604.03191, 2026.
14. This review with the interactive timeline: bushuyeu.com/review-fast-lineage.